

CLIMATE SCIENCE REFERENCE DOCUMENT

Section 1: Global Temperature Trends

The global average temperature has increased by approximately 1.1 degrees Celsius since the pre-industrial era. This warming is primarily due to human activities, particularly the burning of fossil fuels and deforestation.

Section 2: Carbon Dioxide Levels

Atmospheric CO₂ concentration has risen from 280 ppm in pre-industrial times to approximately 420 ppm as of 2024. This represents a 50% increase in atmospheric carbon dioxide.

Section 3: Sea Level Rise

Global mean sea level has risen by approximately 21-24 centimeters since 1880. The rate of sea level rise has accelerated to 3.7 mm per year in recent decades.

Section 4: Arctic Ice

Arctic sea ice extent has declined by approximately 13% per decade since 1979. The Arctic could experience ice-free summers by 2050 under current trends.

Section 5: Greenhouse Gases

The main greenhouse gases are carbon dioxide, methane, nitrous oxide, and fluorinated gases. Methane has 80 times the warming potential of CO₂ over a 20-year period.

Section 6: Renewable Energy

Solar and wind energy now account for approximately 12% of global electricity generation. The cost of solar photovoltaic panels has dropped by 99% since 1976.